

Tutorial 5

PST-1 Bias & Variance of Kernel Density Estimator

$$\hat{p}(x) = \frac{1}{Nh} \sum_{i=1}^N k\left(\frac{x-x_i}{h}\right), \quad D = \{x_i\}_{i=1}^N \text{ samples}$$

h : bandwidth.

$x_i \sim p(x)$ $\forall i$ (iid)
 c.v. x_i is a "sample"

For convenience, let $\tilde{k}(x) = \frac{1}{h} k\left(\frac{x}{h}\right) \Rightarrow \hat{p}(x) = \frac{1}{N} \sum_i \tilde{k}(x-x_i)$

$\uparrow$ scales amplitude (so it integrates to 1)
 $\uparrow$ scales width

a) Mean: $E_{x_1, \dots, x_N} [\hat{p}(x)] = E_{x_1, \dots, x_N} \left[\frac{1}{N} \sum_{i=1}^N \tilde{k}(x-x_i) \right]$

$$= \frac{1}{N} \sum_{i=1}^N E_{x_1, \dots, x_N} [\tilde{k}(x-x_i)]$$

$\Rightarrow E_{x_i}$ $\uparrow$ only depends on x_i

$$= \frac{1}{N} \sum_{i=1}^N E_{x_i} [\tilde{k}(x-x_i)]$$

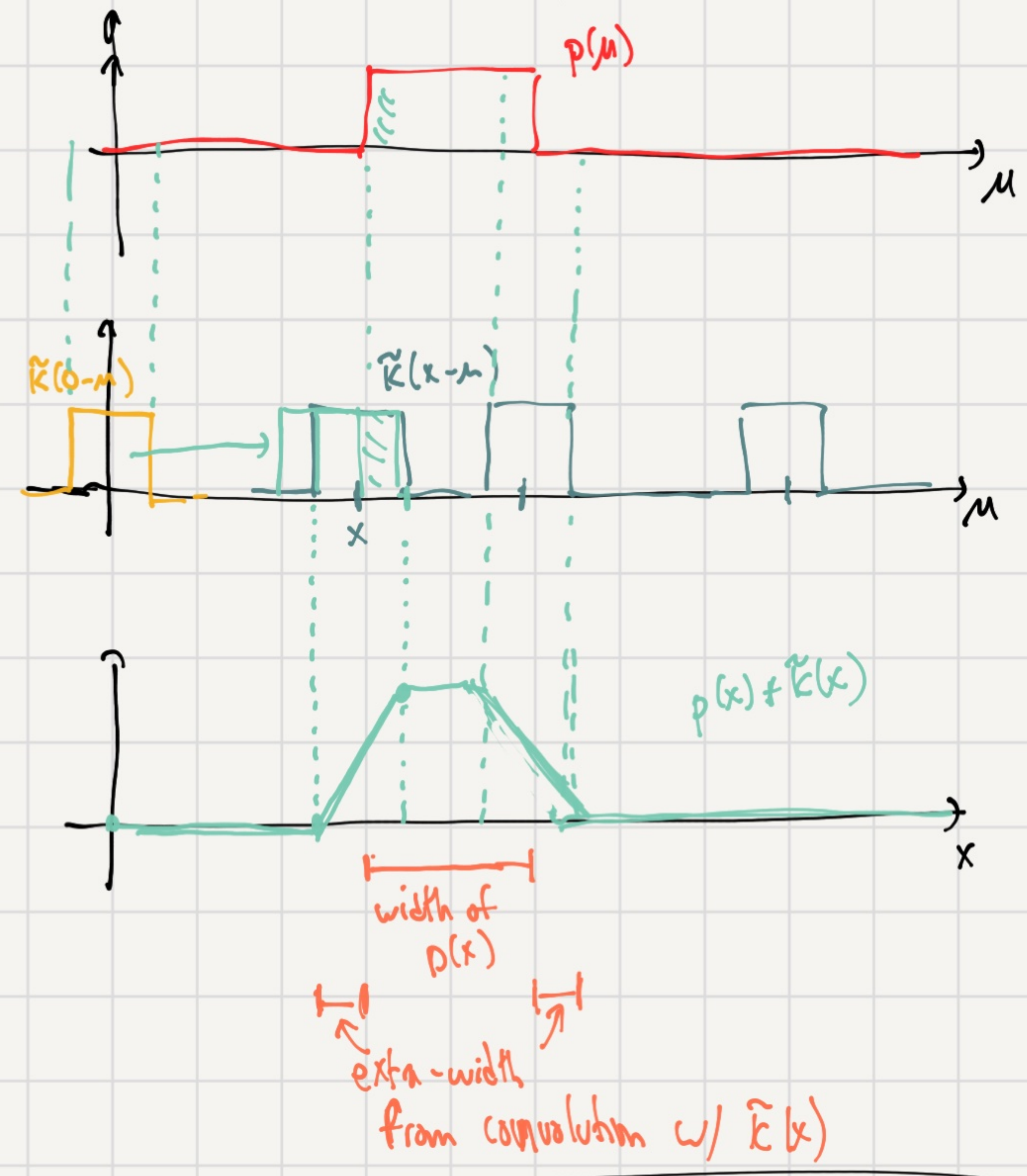
$$= \frac{1}{N} \sum_{i=1}^N \int p(x_i) \tilde{k}(x-x_i) dx_i$$

$\uparrow$ x_i is iid so $p(x_1) = p(x_N) = \dots$
 Same for all i .

$$= \int p(\mu) \tilde{k}(x-\mu) d\mu = p(x) * \tilde{k}(x)$$

$\uparrow$ change x_i to μ . $\uparrow$ convolution

convolution: $p(x) * \tilde{k}(x) = \int p(\mu) \tilde{k}(x-\mu) d\mu$



$E[\hat{p}(x)] = p(x) * \tilde{k}(x) \Leftarrow$ "blurred version of $p(x)$ "
 since $\tilde{k}(x) \geq 0$

Therefore, KDE is only unbiased when
 $E[\hat{p}(x)] = p(x) * \tilde{k}(x) = p(x)$

This only happens when $h=0$, i.e. Dirac delta function
 $\delta(x) = \lim_{h \rightarrow 0} \tilde{k}(x) =$

b) variance.

$$\text{Var}(\hat{p}(x)) = \text{Var}\left(\frac{1}{N} \sum_{i=1}^N \tilde{k}(x-x_i)\right) = \frac{1}{N^2} \text{Var}\left(\sum_{i=1}^N \tilde{k}(x-x_i)\right)$$

$$= \frac{1}{N^2} \sum_{i=1}^N \text{Var}(\tilde{k}(x-x_i))$$

Independence of X_i 's
or variance of sum of r.v.
(PS1-4)

$$= \frac{1}{N} \text{Var}(\tilde{k}(x-x_i))$$

X_i are iid.

$$\leq \frac{1}{N} E[\tilde{k}(x-x_i)^2]$$

$$\text{Var}(X) = E(X^2) - (EX)^2$$

$$\text{Var}(X) \leq E[X^2] \quad \geq 0$$

$$= \frac{1}{N} \int p(x_i) \tilde{k}(x-x_i)^2 dx_i$$

$$\tilde{k}(x-x_i)^2 = \tilde{k}(x-x_i) \tilde{k}(x-x_i) \leq \tilde{k}(x-x_i) \cdot \max \tilde{k}(x-x_i)$$

$$= \tilde{k}(x-x_i) \max \frac{1}{h^d} \tilde{k}\left(\frac{x-x_i}{h}\right)$$

$$= \tilde{k}(x-x_i) \frac{1}{h^d} \max_x \tilde{k}(x)$$

$$\leq \frac{1}{N h^d} \max_x \tilde{k}(x) \int p(x_i) \tilde{k}(x-x_i) dx_i$$

$$\text{Var}(\hat{p}(x)) \leq \frac{1}{N h^d} \left(\max_x \tilde{k}(x)\right) E[\hat{p}(x)]$$

For small variance, we need h large or N large.

h controls the tradeoff b/w bias & variance:

$$\begin{cases} h \rightarrow 0 \Rightarrow \text{bias} \rightarrow 0, \text{var} \rightarrow \infty \\ h \rightarrow \text{large} \Rightarrow \text{bias} \neq 0, \text{var} \rightarrow 0 \end{cases}$$